CRITERION FOR THE NUMBER OF BRANCHES TO BE PRESERVED BY COMPLETION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The map A^h -> A^wedge induces a surjection from the minimal primes of A^wedge to the minimal primes of A^h, and branches(A) is at most branches(A^wedge). invoked at proof lines 31972-31973. Provides the comparison map between the two branch sets.
- Both branch sets are finite, the henselization and the completion being Noetherian. invoked at proof lines 31974-31977. Turns the surjection into a bijectivity criterion.

Source reference. Original lemma 0C28: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31935–31935. Cited at proof lines 31972-31973 for the comparison map and inequality; the conclusion is used inside the counted span.

Source reference. Original lemma 00FR: sources/stacks-chow/tags.txt, source lines 581–581. Cited at proof lines 31974-31975 for the finiteness of the branch sets; the conclusion is used inside the counted span.

Source reference. Original lemma 06LJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10979–10979. Cited at proof lines 31976-31977 for the Noetherianity of A^h and A^wedge; the conclusion is used inside the counted span.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a Noetherian local ring. The number of branches of A is the same as the number of branches of $A^\wedge$ if and only if $\sqrt{\mathfrak{q}A^\wedge}$ is prime for every minimal prime $\mathfrak{q} \subset A^h$ of the henselization.*

Proof. Follows from Lemma 0C28 and the fact that there are only a finite number of branches for both A and $A^\wedge$ by Algebra, Lemma 00FR and the fact that A^h and $A^\wedge$ are Noetherian (Lemma 06LJ). □

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